

Smart Traffic Lights Using Local AI

Varshitha.B.S

Department Computer Science Engineering and Data Science
Ballari Institute of Technology And Management
Visvesvaraya Technological University (VTU), Belagavi
Email:varshithabhari@gmail.com

Abstract—Traditional traffic signal systems generally operate using fixed timing plans or limited sensor information. Such systems may not respond efficiently to changing traffic conditions. This paper proposes a Smart Traffic Light System using Local Artificial Intelligence (AI) to detect traffic conditions and dynamically control signal timings.

The proposed system processes traffic information locally using an edge computing device, reducing dependence on cloud connectivity. The system can use cameras or sensors to estimate traffic volume, occupancy, waiting vehicles, and emergency vehicle presence.

Based on these inputs, a local AI model determines suitable signal phases and timing. The proposed approach focuses on low-cost implementation, real-time decision making, privacy, and reliable operation even when internet connectivity is unavailable.

Index Terms—Smart Traffic Lights, Local AI, Edge Computing, Traffic Management, Computer Vision, Intelligent Transportation System, Emergency Vehicle Detection

I. INTRODUCTION

Traffic congestion is a major problem in urban areas. Conventional traffic lights commonly use predefined timing schedules. These fixed timings cannot always respond to sudden changes in traffic density.

Artificial Intelligence (AI) and computer vision can be used to identify traffic conditions from camera or sensor data. However, cloud-based AI systems require continuous network connectivity and may introduce additional communication delay.

Local AI, also called edge AI, performs data processing close to the source. In a smart traffic signal system, an edge computer can receive camera or sensor information, analyze traffic locally, and provide control decisions to the traffic signal controller.

The proposed system follows the architecture:

Street Cameras/Sensors → Local AI/Edge Computer
→ Traffic Signal Controller → Traffic Lights (1)

An optional cloud connection can be used for data storage, historical analysis, dashboards, and model updates.

II. LITERATURE REVIEW

Research on intelligent traffic signal control has investigated machine learning, reinforcement learning, computer vision, and adaptive control techniques.

Rasheed *et al.* presented a review of deep reinforcement learning approaches for traffic signal control and discussed

the potential of learning-based methods for improving signal control under changing traffic conditions [1].

Gorodokin *et al.* investigated adaptive traffic light control using machine vision and discussed how visual information can be used for traffic-responsive signal control [2].

These studies demonstrate the usefulness of AI and machine vision for traffic management. However, there is still scope for practical low-cost systems that perform AI processing locally and support real-time intersection-level decisions.

III. RESEARCH GAP

Existing research demonstrates the potential of AI-based traffic signal control and machine vision. However, practical deployment can be challenging because of hardware cost, network dependency, computational requirements, and system complexity.

The proposed research focuses on the following gap:

Integration of low-cost local AI with real-time traffic detection and practical emergency-vehicle priority at an intersection.

The objective is to develop a system that can perform important traffic decisions locally without requiring continuous cloud communication.

IV. PROPOSED SYSTEM

The proposed Smart Traffic Light System consists of four major components:

- 1) Traffic cameras or sensors
- 2) Local AI/edge computing device
- 3) Traffic signal controller
- 4) Traffic lights

The cameras or sensors collect information about vehicles approaching the intersection. The local AI system processes the collected information and estimates traffic conditions.

The signal controller receives the decision generated by the local AI system and controls the traffic lights accordingly.

V. SYSTEM ARCHITECTURE

The proposed system architecture is shown below.

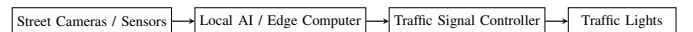


Fig. 1. Proposed architecture of the Smart Traffic Light System.

A. Optional Cloud Connection

Although the main decision-making process is performed locally, an optional cloud connection can be included for:

- Historical traffic data storage
- Monitoring dashboards
- Performance analysis
- Model updates
- Long-term traffic analysis

The cloud is therefore not required for the basic real-time traffic signal operation.

VI. SYSTEM FLOWCHART

The basic operation of the proposed system is represented using the following flowchart.

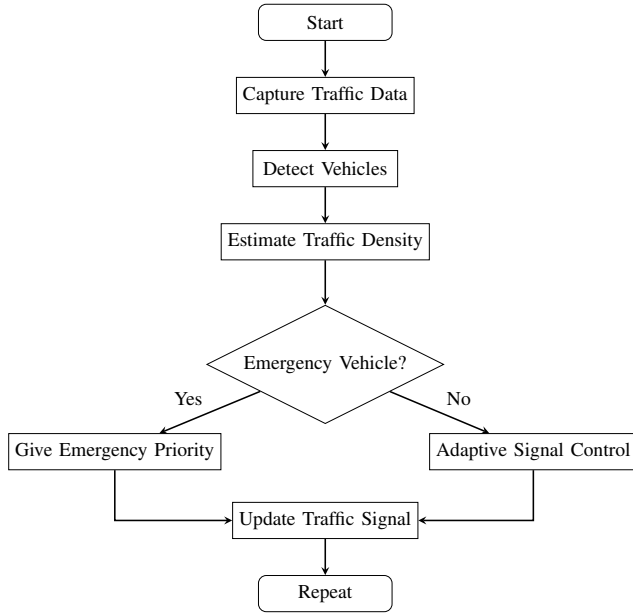


Fig. 2. Flowchart of the proposed local AI traffic control system.

VII. METHODOLOGY

The proposed methodology consists of the following steps.

A. Traffic Data Collection

Traffic data can be collected using cameras or sensors installed near an intersection. The collected information may include:

- Vehicle count
- Lane occupancy
- Waiting vehicles
- Traffic density
- Vehicle movement
- Emergency vehicle presence

B. Local AI Processing

The collected traffic information is processed by a local AI model. Instead of sending every camera frame to a remote server, the edge computer performs the necessary analysis locally.

This reduces network dependency and can improve response time.

C. Traffic Density Estimation

Traffic density can be estimated using vehicle count and lane occupancy.

For lane i , a normalized traffic density can be represented as:

$$D_i = \frac{V_i}{V_{\max}} \quad (2)$$

where:

- V_i is the detected vehicle count in lane i .
- $V_{\max}$ is the maximum reference vehicle count.

The normalized density is restricted to the range $[0, 1]$:

$$D_i^* = \min(1, \max(0, D_i)) \quad (3)$$

D. Adaptive Signal Timing

The green signal duration can be adjusted according to the estimated traffic density.

$$G_i = G_{\min} + D_i^*(G_{\max} - G_{\min}) \quad (4)$$

where:

- G_i is the calculated green time.
- $G_{\min}$ is the minimum green time.
- $G_{\max}$ is the maximum green time.

E. Traffic Feature Vector

The local AI model can use multiple traffic features.

The feature vector for lane i can be represented as:

$$X_i = [V_i \quad O_i \quad W_i \quad A_i] \quad (5)$$

where:

- V_i = vehicle count
- O_i = lane occupancy
- W_i = waiting vehicle count
- A_i = average vehicle speed

For multiple lanes, the traffic information can be represented by a matrix:

$$X = \begin{bmatrix} V_1 & O_1 & W_1 & A_1 \\ V_2 & O_2 & W_2 & A_2 \\ V_3 & O_3 & W_3 & A_3 \\ V_4 & O_4 & W_4 & A_4 \end{bmatrix} \quad (6)$$

F. Local AI Decision

The local AI model receives the traffic feature matrix and generates a traffic signal decision.

$$Y = f(X) \quad (7)$$

where X represents the traffic features and Y represents the resulting signal-control decision.

VIII. LOCAL AI CONTROL ALGORITHM

The basic algorithm can be represented as follows.

Algorithm 1 Local AI Traffic Signal Control

- 1: Capture traffic data
- 2: Detect vehicles using camera/sensors
- 3: Calculate traffic features
- 4: Estimate traffic density
- 5: Check for emergency vehicles
- 6: **if** emergency vehicle detected **then**
- 7: Give priority to the emergency vehicle lane
- 8: **else**
- 9: Calculate adaptive green time
- 10: Update traffic signal
- 11: **end if**
- 12: Repeat the process

IX. HARDWARE AND SOFTWARE REQUIREMENTS

TABLE I
HARDWARE AND SOFTWARE REQUIREMENTS

Category	Requirement
Hardware	Camera/sensors, edge computing device, traffic signal controller, LEDs/traffic lights, power supply
Software	Python, OpenCV, local AI/ML model, microcontroller programming environment
Optional	Cloud storage, dashboard, remote monitoring

X. QUALITATIVE COMPARISON

The proposed local AI approach can be compared with traditional fixed-time traffic signals.

TABLE II
QUALITATIVE COMPARISON

Feature	Fixed-Time	Local AI
Traffic Adaptation	Low	High
Internet Dependency	Low	Low
Real-Time Detection	Limited	Yes
Emergency Priority	Limited	Possible
Local Processing	No	Yes

XI. EXPERIMENTAL SETUP

A prototype can be developed using a camera, an edge computing device, and a traffic signal controller.

The camera provides traffic information to the local AI system. The AI system analyzes the input and generates a signal-control decision.

For experimental evaluation, traffic conditions can be varied using different numbers of vehicles and different lane densities.

XII. RESULTS AND DISCUSSION

The proposed system can be evaluated using parameters such as:

- Average waiting time
- Vehicle throughput
- Signal response time
- Traffic detection accuracy
- Emergency vehicle response

The following table can be used to record experimental results.

TABLE III
EXPERIMENTAL RESULTS

Parameter	Before	After
Average Waiting Time	[Value]	[Value]
Vehicle Throughput	[Value]	[Value]
Detection Accuracy	[Value]	[Value]
Response Time	[Value]	[Value]

The values in Table III should be replaced with actual experimental measurements after testing the prototype.

XIII. PROTOTYPE IMAGE

The developed prototype of the proposed Smart Traffic Light System is shown in Fig. 3.

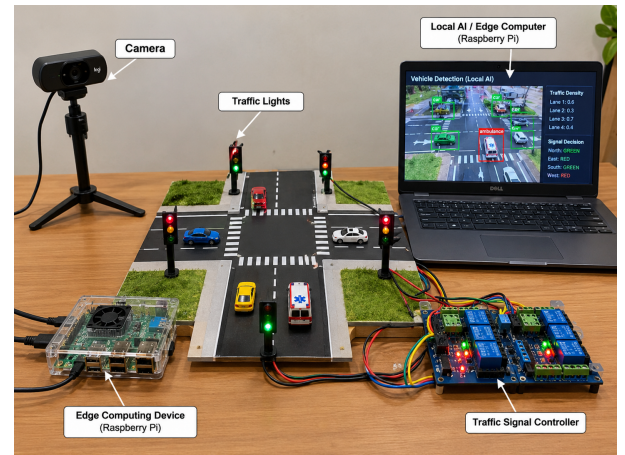


Fig. 3. Prototype of the Smart Traffic Light System.

XIV. POTENTIAL ADVANTAGES

The proposed system has the following potential advantages:

- Real-time traffic adaptation
- Local data processing
- Reduced dependence on internet connectivity
- Faster local decision making
- Potential emergency vehicle priority
- Improved privacy because raw video can be processed locally
- Possibility of low-cost implementation

XV. LIMITATIONS

The system may have several limitations:

- Camera performance may be affected by poor weather or lighting.
- AI models require suitable training data.
- Edge devices have limited computational resources.
- Incorrect vehicle detection can affect signal decisions.
- Prototype testing may not represent all real-world traffic conditions.

XVI. APPLICATIONS

The proposed system can be applied to:

- Urban intersections
- College campuses
- Smart city projects
- Industrial areas
- Emergency corridors
- Small-scale traffic management systems

XVII. FUTURE WORK

Future improvements may include:

- More advanced computer vision models
- Multi-camera traffic monitoring
- Improved emergency vehicle detection
- Reinforcement learning-based signal optimization
- Integration with larger smart-city systems
- Historical traffic prediction
- Remote monitoring dashboards

XVIII. CONCLUSION

This paper presents a Smart Traffic Light System using Local AI. The proposed architecture processes traffic information close to the intersection using an edge computing device.

Traffic conditions can be detected using cameras or sensors, and the local AI model can generate adaptive traffic signal decisions.

The approach reduces dependence on cloud connectivity and provides a foundation for real-time traffic management. Emergency vehicle priority can also be incorporated into the system.

Further prototype development and real-world testing are required to measure the actual performance of the proposed system.

REFERENCES

- [1] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020.
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021.